

Advanced Machine Learning: Augmented Butterfly using Generative Models

Arthur Sophiatti^[2022115599] and Giovanni Faedo^[2025267503]

University of Coimbra, Portugal

Abstract. This work investigates the use of generative data augmentation to improve fine-grained butterfly species classification on a challenging dataset containing 5,199 images distributed across 75 classes. A fixed Convolutional Neural Network (BaselineCNN) is adopted as the classification backbone, following the project requirements. To address class imbalance and data scarcity, the weakest classes are first identified through a baseline performance analysis and subsequently enriched with synthetic samples generated by two conditional generative models: a Conditional Variational Autoencoder (cVAE) and a Conditional Generative Adversarial Network (cGAN).

A confidence-based Top-K filtering strategy is introduced to select only the most reliable generated samples before their integration into the training set. The effectiveness of the proposed augmentation framework is evaluated through classification accuracy and macro F1-score. Experimental results show a progressive improvement from 45.0% accuracy (Macro F1 = 0.418) for the baseline model to 50.5% (Macro F1 = 0.460) using traditional augmentation. Generative augmentation further improves performance, reaching 58.3% accuracy (Macro F1 = 0.559) with the cGAN and 63.8% accuracy (Macro F1 = 0.618) with the cVAE. These results demonstrate that targeted generative augmentation can substantially improve classification performance in highly imbalanced, small-data scenarios, with the cVAE providing the most effective synthetic samples for this task.

Keywords: Generative Data Augmentation · Conditional Variational Autoencoder · Conditional GAN · Butterfly Classification · Class Imbalance

1 Introduction

The automated classification of biological organisms via computer vision has emerged as an important tool for ecological monitoring, biodiversity preservation, and automated field taxonomy [2]. Among insects, butterfly species recognition represents a challenging fine-grained classification task due to subtle inter-species morphological differences, high intra-class variability, and the presence of complex natural backgrounds.

This research addresses the classification of 75 butterfly species using deep learning techniques. Models trained on such biological datasets often suffer from

poor generalization due to two major challenges: limited data availability and severe class imbalance. To address these issues, this study investigates whether targeted generative augmentation can improve the performance of a fixed classification backbone under highly imbalanced conditions.

The proposed methodology begins with the identification of the weakest classes through a baseline performance analysis. Three training configurations are then evaluated: a baseline classifier trained on the original dataset, a version enhanced with traditional geometric augmentation, and two conditional generative approaches based on a Conditional Variational Autoencoder (cVAE) and a Conditional Generative Adversarial Network (cGAN). Synthetic samples are generated exclusively for the weakest categories and subsequently integrated into the training set.

The main contribution of this work is the systematic evaluation of targeted generative augmentation for minority butterfly classes. Rather than augmenting the entire dataset uniformly, synthetic data generation is focused on the categories that most severely limit classification performance. Furthermore, a confidence-based Top-K filtering strategy is introduced to retain only the most reliable generated samples before their integration into the training pipeline.

The remainder of this document is organized as follows. Section 2 describes the dataset, preprocessing pipeline, and identification of weak classes. Section 3 presents the classification backbone and the generative architectures. Section 4 details the experimental setup and evaluation metrics. Section 5 discusses the empirical results, while Section 6 concludes the work and outlines future research directions.

2 Dataset and Preprocessing

The dataset utilized in this study comprises 5,199 images distributed across 75 fine-grained categories of butterfly species. All images were resized to 64×64 pixels and converted into RGB format before training. To preserve the original class distribution across all subsets, a stratified split was performed, allocating 70% of the data to training, 15% to validation, and 15% to testing. During classifier training, geometric augmentation techniques, including random horizontal flips, vertical flips, and rotations, were applied exclusively to the training set. Fine-grained visual categorization (FGVC) in this biological domain represents an inherently non-trivial computer vision task. As illustrated in Fig. 1, the dataset is characterized by low inter-class variance—where distinct species display near-identical morphological wing contours and venation patterns—and high intra-class variance, caused by severe clutter, variable lighting, and arbitrary camera poses in natural habitats.

2.1 Identification of Weak Classes

The original dataset exhibits two interconnected constraints: a limited total sample size and a pronounced class imbalance across the 75 butterfly species. Rather



Fig. 1. Sample images from the dataset, illustrating the complex backgrounds, varying poses, and subtle inter-class differences among the butterfly species.

than applying a blind, global oversampling strategy that could dilute the model’s focus or waste computational resources, we established a targeted probing phase. First, we trained our baseline Convolutional Neural Network (CNN) from scratch utilizing exclusively the unaltered, original dataset. Once training converged, we evaluated the model to map the per-class performance on the test set for each individual species.

Through this quantitative probe, we precisely isolated the *10 worst-performing* categories (referred to as **”weak classes”**). These specific minority classes displayed critical accuracy drops compared to the rest of the dataset. This performance degradation was primarily caused by an extreme scarcity of initial training images and severe inter-class similarity, as the morphological wing patterns of these species heavily overlapped with others, leading to classifier confusion. Consequently, we chose to restrict our generative expansion and data augmentation framework exclusively to these 10 struggling classes to directly fortify the model’s weakest decision boundaries. This targeted strategy was adopted to maximize the impact of synthetic augmentation while avoiding the introduction of unnecessary synthetic samples into classes that were already adequately represented. By concentrating the generative effort on the weakest categories, the augmentation budget could be allocated where it was expected to produce the largest improvement in classification performance.

The quantitative metrics in Table 1 expose the critical vulnerabilities of the baseline classifier. For the first eight species, the model yields a Precision, Recall, and F1-score of exactly 0.00. This behavior suggests that the network failed to learn sufficiently discriminative representations for these categories, likely due to the combined effects of severe class imbalance and strong visual similarity between species.

The *Samples* column highlights the root cause: an acute data-scarce regime with only 9 to 11 instances per class. For *Mestra* and *African Giant Swallowtail*,

the low Recall (0.10 and 0.11) indicates a massive rate of False Negatives, with the network missing approximately 90% of true targets. These results provide the mathematical justification for our generative approach: injecting synthetic variance via cVAE and cGAN is essential to reconstruct these missing decision boundaries.

The identification of weak classes establishes the central hypothesis of this work: improving the representation of minority classes through synthetic sample generation should lead to stronger decision boundaries and improved overall classification performance. The following sections investigate whether conditional generative models can effectively provide such additional training data.

Table 1. Quantitative performance metrics (Precision, Recall, F1-score) of the 10 isolated weak classes under the baseline CNN framework on the Test Set.

Class ID	Species Name	Precision	Recall	F1-score	Samples
10	BLUE MORPHO	0.00	0.00	0.00	9
16	CHESTNUT	0.00	0.00	0.00	10
25	DANAID EGGFLY	0.00	0.00	0.00	11
38	LARGE MARBLE	0.00	0.00	0.00	10
66	SOUTHERN DOGFACE	0.00	0.00	0.00	10
68	TROPICAL LEAFWING	0.00	0.00	0.00	10
11	BLUE SPOTTED CROW	0.166667	0.1	0.125	10
64	SLEEPY ORANGE	0.333333	0.0769231	0.125	13
52	PINE WHITE	0.2	0.1	0.133333	10
22	COPPER TAIL	0.5	0.0909091	0.153846	11

3 Solution Approach

The technical framework engineered in this study addresses data scarcity and severe class imbalance through a two-fold pipeline: a robust convolutional classification backbone trained from scratch, paired with a targeted, label-conditioned generative expansion framework. To enforce specialized visual synthesis, both the Conditional Variational Autoencoder (cVAE) and the Conditional Generative Adversarial Network (cGAN) are mathematically conditioned on the target class labels.

3.1 Classification Backbone: BaselineCNN

To handle the uniform 64×64 RGB pixel spatial dimensions, a modular Convolutional Neural Network (CNN) was developed from scratch. The network is architected to extract structural spatial hierarchies, moving progressively from

low-level edge detection to high-frequency textural patterns characteristic of biological wing venation.

The backbone consists of three sequential convolutional blocks. Each individual block integrates a 2D convolutional layer with a 3×3 kernel, paired with a Batch Normalization layer to stabilize internal covariate shift, a Rectified Linear Unit (ReLU) activation function, and a 2×2 Max-Pooling layer for spatial downsampling. The number of feature maps doubles symmetrically across the blocks, scaling from 64 to 128, and terminating at 256 channels. A Global Average Pooling layer reduces the spatial dimensions to 1×1 . The resulting 256-dimensional feature vector is then passed through three fully connected layers ($256 \rightarrow 512 \rightarrow 1024 \rightarrow 2048$), each followed by ReLU activation and 50% Dropout, before a final linear projection to the $C = 75$ output classes. This architecture was kept unchanged throughout all experiments in order to isolate the impact of the proposed data augmentation strategies.

3.2 Deep Generative Frameworks

Rather than applying a generic global oversampling mechanism, the generative modules are strictly conditioned on the isolated weak classes to selectively densify minority distributions.

Conditional Variational Autoencoder (cVAE) The cVAE was selected as a representative latent-variable generative model capable of producing class-conditioned samples while maintaining a structured latent space [3]. The framework consists of two parameterized sub-networks:

- **The Encoder (E_ϕ):** Maps the flattened pixel vector of a real image concatenated with its corresponding one-hot encoded class label vector y to the parameters of a latent Gaussian distribution, specifically the mean (μ) and the log-variance ($\log \sigma^2$).
- **The Reparameterization Trick:** To preserve end-to-end backpropagation through the stochastic layer, the continuous latent vector z is sampled via:

$$z = \mu + \epsilon \odot \sigma, \quad \text{where } \epsilon \sim \mathcal{N}(0, I) \quad (1)$$
- **The Decoder (D_θ):** Accepts the latent variable z concatenated with the conditioning vector y , mapping the combined representations back into the original pixel space.

The entire architecture is optimized end-to-end by maximizing the Evidence Lower Bound (ELBO), which balances a Mean Squared Error (MSE) Reconstruction Loss with a Kullback-Leibler (KL) Divergence regularization term to align the latent distribution with a standard Gaussian prior:

$$\mathcal{L}_{\text{cVAE}}(\theta, \phi) = E_{q_\phi(z|x, y)}[\log p_\theta(x|z, y)] - D_{\text{KL}}(q_\phi(z|x, y) \parallel p(z)) \quad (2)$$

The training objective follows the standard cVAE formulation [3]. In practice, a small weighting coefficient ($\beta = 0.001$) was applied to the KL-divergence term, resulting in a beta-cVAE variant [4].

Conditional Generative Adversarial Network (cGAN) Generative Adversarial Networks were originally introduced by Goodfellow et al. [5]. The cGAN addresses synthesis via a game-theoretic minimax optimization, where both the Generator (G) and the Discriminator (D) are conditioned on the class label y , following the formulation proposed by Mirza and Osindero [6]:

- **The Generator (G):** Maps a stochastic noise vector $z \sim \mathcal{N}(0, I)$ paired with the class target embedding to synthesize a realistic image $\hat{x} = G(z|y)$, projecting pixel values into the $[-1, 1]$ range via a Hyperbolic Tangent ($\tanh$) activation.
- **The Discriminator (D):** Evaluates whether a given input sample is an authentic biological image or a synthetic artifact generated by G . It is optimized using a Binary Cross-Entropy (BCE) objective function conditioned on the matching target y :

$$\min_G \max_D \mathcal{L}_{\text{cGAN}}(D, G) = E_{x \sim p_{\text{data}}} [\log D(x|y)] + E_{z \sim p_z} [\log(1 - D(G(z|y)|y))] \quad (3)$$

3.3 Confidence-Driven Top-K Filtering

Not all generated samples contribute positively to classification performance. Consequently, a confidence-based filtering strategy was introduced to retain only the most reliable synthetic samples before dataset integration.

A primary architectural contribution of this work is the implementation of a post-generation *Strategic Top-K Selection Filter* designed to mitigate low-quality generations and potential mode-collapse artifacts. The synthesized data distributions undergo a automated quality verification cycle:

1. **Normalization Realignment:** Samples synthesized via the cGAN are rescaled from $[1, 1]$ to $[0, 1]$, while cVAE outputs are already normalized in $[0, 1]$ through a Sigmoid activation.
2. **Confidence Probing:** The initial baseline CNN model (trained strictly on pure real data) acts as an external frozen critic. It processes the massive pool of raw synthetic generations to compute the exact softmax probability for the correct ground-truth class target y .
3. **Elite Selection:** For each weak class, the generations are sorted by confidence, and only the elite top $K = 80$ samples displaying the highest correct-class probability are retained, filtering out structural anomalies.
4. **Deterministic Concatenation:** The filtered high-fidelity tensors are triplicated via a multiplication factor of $3\times$, yielding $K \times 3 = 240$ synthetic samples per weak class. This expansion factor was selected to balance the severe under-representation of the 10 weak classes (9–11 real samples each) against the risk of dataset poisoning from low-quality generations. Combined with geometric augmentation inside a custom `AugmentedTensorDataset` wrapper, this establishes a training distribution where minority classes reach a density comparable to majority classes, without overwhelming the real data distribution.

4 Experimental Setup

To ensure a rigorous and unbiased comparison across all generative and classification architectures, all experiments were conducted using the PyTorch framework, accelerated by a CUDA-enabled GPU (Google Colab T4). The complete grid of tested configurations is summarized in Table 2.

Table 2. Systematic hyperparameter configuration and optimization profile for the generative modules and the classification backbone.

Pipeline Component	Hyperparameter Description	Selected Value
Data Setup	Spatial Resolution (Resized)	64×64 pixels
	Mini-batch Size	32
	Train / Val / Test Split Ratio	70% / 15% / 15%
BaselineCNN	Optimization Algorithm	Adam
	Base Learning Rate (η)	10^{-3}
	Dense Layer Dropout Rate	50%
	Loss Function	Cross-Entropy
Generative Phases	cVAE Learning Rate	5×10^{-4}
	cGAN Learning Rate	2×10^{-4}
	Latent Space Dimensions (z)	100
	cVAE Weight Decay (L_2 Penalty)	10^{-4}
Top-K Filtering	Confidence Cut-off Threshold (K)	80 samples/class
	Dataset Multiplication Factor	$3\times$
	Targeted Classes	10 weak classes

4.1 Data Augmentation Pipeline

To combat the severe overfitting inherent in fine-grained classification with limited samples, we integrated a spatial augmentation pipeline into the training loop of the classification backbone. Rather than utilizing aggressive pixel-level distortions that could destroy fine-grained biological features, we employed conservative geometric transformations. Specifically, the pipeline applies random horizontal flips ($p = 0.5$), random vertical flips ($p = 0.3$), and random rotations (up to 30 degrees). These transformations simulate varying camera angles and natural butterfly poses without corrupting the underlying morphological textures, providing a robust defense against memorization of the training set. The same augmentation pipeline was applied across all classification experiments to ensure that performance differences could be attributed to the generated samples rather than to different preprocessing strategies.

4.2 Training Dynamics and Optimization

The classification networks were optimized using the Adam optimizer, incorporating a weight decay of 10^{-4} to explicitly penalize extreme weight magnitudes and improve generalization. To dynamically adapt the learning rate during training, a `ReduceLROnPlateau` scheduler was deployed. This scheduler monitors the validation loss and halves the learning rate (factor of 0.5) if no improvement is observed for 5 consecutive epochs, allowing the optimizer to make finer adjustments as it approaches a local minimum.

To ensure optimal weight retention and prevent overfitting, an Early Stopping callback was configured. The training process was halted if the validation loss stagnated for 5 consecutive epochs, and the model state corresponding to the lowest validation loss was restored for final evaluation.

4.3 Evaluation Metrics

Model performance was quantitatively assessed using a macro-averaged formulation of Precision, Recall, and the F1-score. Macro-averaged metrics were preferred over micro-averaged alternatives because they assign equal importance to all classes, regardless of their frequency. This choice is particularly relevant in the present study, where the primary objective is to improve the performance of severely underrepresented butterfly species. Using a macro-average ensures that the minority weak classes, which are disproportionately affected by data scarcity, contribute equally to the final metric score as the majority classes. Finally, the definitive benchmarking of the models' generalized capabilities was conducted on an unseen hold-out test set through the official Kaggle competition platform, relying strictly on global classification accuracy.

To assess the quality of the generated images, three standard metrics were computed on 2,000 synthetic samples: the Fréchet Inception Distance (FID), the Inception Score (IS), and the Structural Similarity Index (SSIM). As reported in Table 3, the cGAN achieves superior FID and IS, confirming that adversarial training produces more realistic and diverse images. The cVAE obtains a higher SSIM, consistent with its encoder-decoder architecture that reconstructs images anchored to real observations. Both FID values remain high in absolute terms, reflecting the difficulty of generating fine-grained butterfly textures at 64×64 resolution.

It is important to note that generative quality metrics and classification performance evaluate different objectives. Consequently, a model achieving superior FID or Inception Score values is not necessarily expected to produce the largest improvement in downstream classification accuracy.

5 Analysis of the Results

The experimental evaluation was designed to measure the progressive impact of generative data augmentation on the classification performance, particularly

Table 3. Generative evaluation metrics for cVAE and cGAN on 2,000 synthetic samples per model.

Model	FID	IS	SSIM
cVAE	320.7277	1.3955 ± 0.0403	0.1756
cGAN	280.2109	2.7573 ± 0.0601	0.0560

focusing on the mitigation of severe class imbalance. The results, summarized in Table 4 and visually depicted in Figure 2, demonstrate a clear evolutionary improvement across the three tested paradigms.

Table 4. Test Accuracy across the three experimental paradigms, highlighting the progressive impact of generative data augmentation.

Model Architecture	Data Strategy	Test Accuracy	Macro F1-Score
Baseline CNN	Flips + Rotations	45.0%	0.418
Baseline CNN + Simple Aug.	Real + Simple aug.	50.5%	0.460
BaselineCNN + cVAE	Real + Simple aug. + Synthetic cVAE	63.8%	0.618
BaselineCNN + cGAN	Real + Simple aug. + Synthetic cGAN	58.3%	0.559

5.1 Baseline Performance and the Data Constraint

The baseline classifier achieved an overall accuracy of 45.0% and a macro F1-score of 0.418. A detailed class-wise analysis revealed substantial performance disparities across categories, with several minority classes exhibiting near-zero recall and F1-scores. These results confirmed the presence of severe class imbalance and motivated the adoption of targeted augmentation strategies.

5.2 Traditional Augmentation

Applying standard geometric augmentation improved the overall accuracy to 50.5% and increased the macro F1-score to 0.460. This result suggests that additional viewpoint and orientation variability improves the generalization capability of the classifier. However, the improvement remained limited for the weakest categories, indicating that conventional augmentation alone could not fully compensate for the scarcity of training samples.

5.3 Generative Augmentation

The introduction of conditional generative models produced the largest performance improvements observed in this study. The cGAN-based augmentation

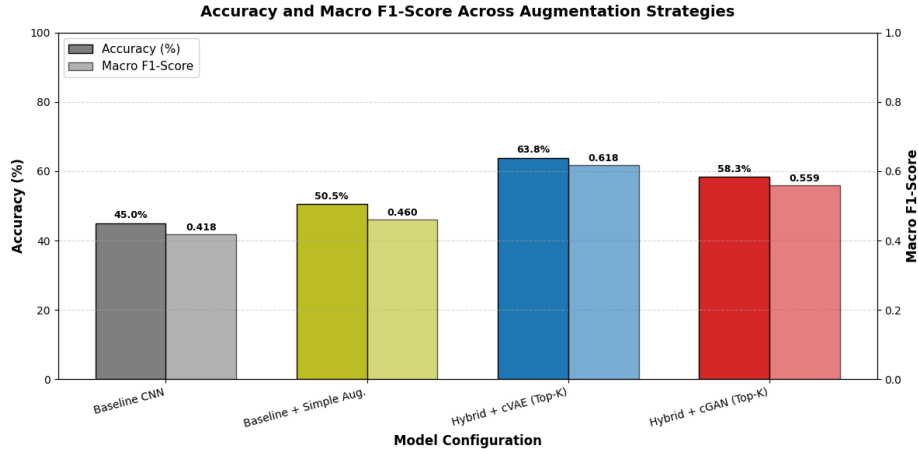


Fig. 2. Comparative bar chart illustrating the progressive improvement in test accuracy across the Baseline, Baseline with Simple Augmentation, cVAE and cGAN models

strategy increased the classification accuracy from 45.0% to 58.3%, while also improving the macro F1-score from 0.418 to 0.559. An even stronger improvement was achieved using the cVAE framework, which obtained the best overall results with 63.8% accuracy and a macro F1-score of 0.618.

Interestingly, the cVAE outperformed the cGAN despite the latter being specifically designed to generate highly realistic images. This result suggests that visual realism alone is not sufficient to maximize downstream classification performance. In our setting, the cVAE likely produced synthetic samples that better preserved the underlying class characteristics, resulting in more informative training examples for the classifier. Overall, both generative approaches substantially outperformed traditional augmentation, confirming the effectiveness of targeted synthetic data generation for minority butterfly classes.

6 Conclusion

This work investigated the effectiveness of targeted generative augmentation for fine-grained butterfly species classification under severe class imbalance. Weak classes were identified through a baseline CNN and selectively enriched using synthetic samples generated by a Conditional Variational Autoencoder (cVAE) and a Conditional Generative Adversarial Network (cGAN). A confidence-based Top-K filtering strategy was employed to retain only the most reliable generated samples before their integration into the training set.

The results showed that generative augmentation substantially outperformed both the baseline model and traditional geometric augmentation. While the cGAN achieved significant improvements, the cVAE obtained the best overall performance, reaching 63.8% accuracy and a macro F1-score of 0.618. These

findings suggest that, in highly imbalanced and data-constrained scenarios, preserving the underlying class distribution may be more beneficial for classifier training than maximizing visual realism alone.

Future work may explore more advanced generative models and alternative sample selection strategies to further enhance classification performance. The complete implementation and experimental pipeline are publicly available in the project repository [1].

Declaration of AI Usage

During the preparation of this work, the authors utilized Large Language Models (LLMs) exclusively as formatting and copy-editing assistants. Specifically, AI was employed to assist with L^AT_EX typesetting, structural organization, and translating rough drafts into formal academic English.

The authors rigorously reviewed, edited, and validated all generated text. The core scientific contributions of this paper—including the conceptualization of the problem, the writing of the Python code, the experimental design, the hyperparameter tuning, the generation of graphs, and the critical analysis of the results—were entirely performed by the human authors. The authors take full responsibility for the scientific integrity, accuracy, and originality of the content presented in this document, and are fully prepared to demonstrate a deep understanding of all technical components during the project defense.

References

1. AML More Butterfly: Project Source Code and Data. Arthur Sophiatti and Giovanni Faedo. (2026) <https://github.com/Tuco711/AML.More.Butterflies>
2. Urooj, S., et al.: Automated Classification of Butterfly Species in the Hyderabad Region Using Deep Learning and Image Recognition. *International Journal of Intelligent Systems and Applications in Engineering* **13**(4), 589–602 (2025). www.researchgate.net/publication/395935134
3. Kingma, D.P., Welling, M.: Auto-Encoding Variational Bayes. *International Conference on Learning Representations (ICLR)* (2014). <https://arxiv.org/abs/1312.6114>
4. Higgins, I., Matthey, L., Pal, A., Burgess, C., Glorot, X., Botvinick, M., Mohamed, S., Lerchner, A.: beta-VAE: Learning Basic Visual Concepts with a Constrained Variational Framework. *International Conference on Learning Representations (ICLR)* (2017). <https://openreview.net/forum?id=Sy2fzU9gl>
5. Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y.: Generative Adversarial Nets. *Advances in Neural Information Processing Systems (NeurIPS)* 27 (2014). <https://arxiv.org/abs/1406.2661>
6. Mirza, M., Osindero, S.: Conditional Generative Adversarial Nets. *arXiv preprint arXiv:1411.1784* (2014). <https://arxiv.org/abs/1411.1784>